

- Patients should follow the same titration schedule when they are naive to methylphenidate or are switching from another formulation



Dosing Tips

- Clinical duration of action often differs from pharmacokinetic half-life
- Taking oral formulations with food may delay peak actions for 2–3 hours
- Do not substitute methylphenidate products on a mg-per-mg basis due to differing methylphenidate base compositions and pharmacokinetic profiles
- * Immediate-release formulations (Ritalin, Methylin, generic methylphenidate) have 2–4 hour durations of clinical action
- * Sustained-release formulations such as Methylin ER, Ritalin SR, Metadate ER, and generic methylphenidate sustained-release all have approximately 4–6 hour durations of clinical action, which for most patients is generally not long enough for once daily dosing in the morning and thus generally requires lunchtime dosing at school
- * Sustained-release Metadate CD has an early peak and an 8-hour duration of action
- * Sustained-release Ritalin LA also has an early peak and an 8-hour duration of action, with 2 pulses (immediate and after 4 hours)

* Sustained-release Concerta trilayer tablet, orally disintegrating tablet Cotempla XRODT, Quillivant XR, and Aptensio XR have 12-hour durations of action

- Orally disintegrating tablet Cotempla-XR-ODT has a 12-hour duration of action
- Most sustained-release formulations (especially Concerta, Metadate CD, and Ritalin LA) should not be chewed but rather should only be swallowed whole
- QuilliChew ER is a chewable tablet and can be taken with or without food
- Jornay PM can be taken with or without food, but should be consistent; the capsule can be opened and the contents sprinkled onto applesauce
- Jornay PM: if the evening dose is missed and not remembered until the next morning, then the patient should wait until their next scheduled evening dose
- Adhansia XR can be taken with or without food; the capsule can be opened and the contents sprinkled onto applesauce or yogurt; if mixture is not consumed within 10 minutes it should be discarded; 45 mg capsule contains tartrazine, which may cause allergic reactions, particularly in patients allergic to aspirin
- Adhansia XR: if the morning dose is missed, then the patient should wait until their next scheduled morning dose

* Newer sustained-release formulations have a sufficiently long duration of clinical action to eliminate the need for a lunchtime dosing if taken in the morning

* This innovation can be an important practical element in stimulant utilization, eliminating the hassle and pragmatic difficulties of lunchtime dosing at school, including storage problems, potential diversion, and the need for a medical professional to supervise dosing away from home

- Off-label uses are dosed the same as for ADHD

* May be possible to dose only during the school week for some ADHD patients

* May be able to give drug holidays over the summer in order to reassess therapeutic utility and effects on growth and to allow catch-up from any growth suppression as well as to assess any other side effects and the need to reinstitute stimulant treatment for the next school term

- However, most studies show that parental height is what determines a patient's final height, and that most children/adolescents taking stimulants reach their expected height, just more slowly than children/adolescents not exposed to stimulants

- Avoid dosing late in the day because of the risk of insomnia

- Concerta tablet does not change shape in the gastrointestinal tract and generally should not be used in patients with gastrointestinal narrowing because of the risk of intestinal obstruction

- Side effects are generally dose-related

- Transdermal patch should be applied to dry, intact skin on the hip
- New application site should be selected for each day; only one patch should be applied at a time; patches should not be cut
- Avoid touching the exposed (sticky) side of the patch, and after application, wash hands with soap and water; do not touch eyes until after hands have been washed
- Heat can increase the amount of methylphenidate absorbed from the transdermal patch, so patients should avoid exposing the application site to external source of direct heat (e.g., heating pads, prolonged direct sunlight)
- If a patch comes off a new patch may be applied at a different site; total daily wear time should remain 9 hours regardless of number of patches used
- Early removal of transdermal patch can be useful to terminate drug action when desired

Overdose

- Vomiting, tremor, coma, convulsion, hyperreflexia, euphoria, confusion, hallucination, tachycardia, flushing, palpitations, sweating, hyperreflexia, hypertension, arrhythmia, mydriasis

Long-Term Use

- Often used long-term for ADHD when ongoing monitoring documents continued efficacy

- Dependence and/or abuse may develop
- Tolerance to therapeutic effects may develop in some patients
- Long-term stimulant use may be associated with growth suppression in children (controversial)
- Periodic monitoring of weight, blood pressure, CBC, platelet counts, and liver function may be prudent

Habit Forming

- High abuse potential, Schedule II drug
- Patients may develop tolerance, psychological dependence

How to Stop

- Taper to avoid withdrawal effects
- Withdrawal following chronic therapeutic use may unmask symptoms of the underlying disorder and may require follow-up and reinstitution of treatment
- Careful supervision is required during withdrawal from abusive use since severe depression may occur

Pharmacokinetics

- Average half-life in adults is 3.5 hours (1.3–7.7 hours)
- Average half-life in children is 2.5 hours (1.5–5 hours)

- First-pass metabolism is not extensive with transdermal dosing, thus resulting in notably higher exposure to methylphenidate and lower exposure to metabolites as compared to oral dosing

Drug Interactions

- May affect blood pressure and should be used cautiously with agents used to control blood pressure
- May inhibit metabolism of SSRIs, anticonvulsants (phenobarbital, phenytoin, primidone), TCA, and coumarin anticoagulants, requiring downward dosage adjustments of these drugs
- Serious adverse effects may occur if combined with clonidine (controversial)
- Use with MAOIs, including within 14 days of MAOI use, is not advised, but this can sometimes be considered by experts who monitor depressed patients closely when other treatment options for depression fail
- CNS and cardiovascular actions of d,l-methylphenidate could theoretically be enhanced by combination with agents that block norepinephrine reuptake, such as the TCAs desipramine or protriptyline, venlafaxine, duloxetine, atomoxetine, milnacipran, and reboxetine
- Theoretically, antipsychotics should inhibit the stimulatory effects of d,l-methylphenidate

- Theoretically, d,l-methylphenidate could inhibit the antipsychotic actions of antipsychotics
- Theoretically, d,l-methylphenidate could inhibit the mood-stabilizing actions of atypical antipsychotics in some patients
- Combination of d,l-methylphenidate with mood stabilizers (lithium, anticonvulsants, atypical antipsychotics) is generally something for experts only, when monitoring patients closely and when other options fail

Other Warnings/Precautions

- Use with caution in patients with any degree of hypertension, hyperthyroidism, or history of drug abuse
- Children who are not growing or gaining weight should stop treatment, at least temporarily
- May worsen motor and phonic tics
- May worsen symptoms of thought disorder and behavioral disturbance in psychotic patients
- Stimulants have a high potential for abuse and must be used with caution in anyone with a current or past history of substance abuse or alcoholism or in emotionally unstable patients
- Administration of stimulants for prolonged periods of time should be avoided whenever possible or done only with close monitoring, as it may lead to marked tolerance and drug dependence, including

psychological dependence with varying degrees of abnormal behavior

- Particular attention should be paid to the possibility of subjects obtaining stimulants for nontherapeutic use or distribution to others and the drugs should in general be prescribed sparingly with documentation of appropriate use
- Usual dosing has been associated with sudden death in children with structural cardiac abnormalities
- Not an appropriate first-line treatment for depression or for normal fatigue
- May lower the seizure threshold
- Emergence or worsening of activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of a mood stabilizer and/or discontinuation of d,l-methylphenidate
- Permanent skin color loss, known as chemical leukoderma, may occur with use of the transdermal Daytrana patch; patients should be advised to watch for signs of skin color changes and if they occur alternative treatment options should be considered
- Certain transdermal patches containing even small traces of aluminum or other metals in the adhesive backing can cause skin burns if worn during MRI, so warn patients taking the transdermal

formulation about this possibility and advise them to disclose this information if they need an MRI

Do Not Use

- If patient has extreme anxiety or agitation
- If patient has motor tics or Tourette's syndrome or if there is a family history of Tourette's, unless administered by an expert in cases when the potential benefits for ADHD outweigh the risks of worsening tics
- Should generally not be administered with an MAOI, including within 14 days of MAOI use, except in heroic circumstances and by an expert
- If patient has glaucoma
- If patient has structural cardiac abnormalities
- If there is a proven allergy to methylphenidate

Special Populations

Renal Impairment

- No dose adjustment necessary

Hepatic Impairment

- No dose adjustment necessary

Cardiac Impairment

- Use with caution, particularly in patients with recent myocardial infarction or other conditions that could be negatively affected by increased blood pressure
- Do not use in patients with structural cardiac abnormalities

Elderly

- Some patients may tolerate lower doses better



Children and Adolescents

- Safety and efficacy not established in children under age 6
- Use in young children should be reserved for the expert
- Methylphenidate has acute effects on growth hormone; long-term effects are unknown but weight and height should be monitored during long-term treatment
- Half-life and duration of clinical action tend to be shorter in younger children
- Usual dosing has been associated with sudden death in children with structural cardiac abnormalities
- American Heart Association recommends ECG prior to initiating stimulant treatment in children, although not all experts agree



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Infants whose mothers took methylphenidate during pregnancy may experience withdrawal symptoms
- Racemic methylphenidate has been shown to have teratogenic effects in rabbits when given in doses of 200 mg/kg/day throughout organogenesis
- Use in women of childbearing potential requires weighing potential benefits to the mother against potential risks to the fetus
- * For ADHD patients, methylphenidate should generally be discontinued before anticipated pregnancies

Breast Feeding

- Unknown if methylphenidate is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- If infant shows signs of irritability, drug may need to be discontinued

The Art of Psychopharmacology

Potential Advantages

- Established long-term efficacy as a first-line treatment for ADHD
- Multiple options for drug delivery, peak actions, and duration of action

Potential Disadvantages

- Patients with current or past substance abuse
- Patients with current or past bipolar disorder or psychosis
- Patients with anorexia
- Patients with insomnia

Primary Target Symptoms

- Concentration, attention span
- Motor hyperactivity
- Impulsiveness
- Physical and mental fatigue
- Daytime sleepiness
- Depression

* May be useful for treatment of depressive symptoms in medically ill elderly patients

* May be useful for treatment of post-stroke depression

* A classical augmentation strategy for treatment-refractory depression

* Specifically, may be useful for treatment of cognitive dysfunction and fatigue as residual symptoms of major depressive disorder unresponsive to multiple prior treatments

* May also be useful for the treatment of cognitive impairment, depressive symptoms, and severe fatigue in patients with HIV infection and in cancer patients

- Can be used to potentiate opioid analgesia and reduce sedation, particularly in end-of-life management
- Atypical antipsychotics may be useful in treating stimulant or psychotic consequences of overdose
- Some patients respond to or tolerate methylphenidate better than amphetamine and vice versa
- Drug abuse may actually be lower in ADHD adolescents treated with stimulants than in ADHD adolescents who are not treated
- Older sustained-release technologies for methylphenidate were not significant advances over immediate-release methylphenidate because they did not eliminate the need for lunchtime dosing or allow once daily administration

* Newer sustained-release technologies are truly once a day dosing systems

* Metadate CD and Ritalin LA are somewhat similar to each other, both with an early peak and duration of action of about 8 hours

* Concerta has less of an early peak but a longer duration of action (up to 12 hours)

* Concerta trilayer tablet consists of 3 compartments (2 containing drug, 1 a “push” compartment) and an orifice at the head of the first drug compartment; water fills the push compartment and gradually pushes drug up and out of the tablet through the orifice

* Concerta may be preferable for those ADHD patients who work in the evening or do homework up to 12 hours after morning dosing

* Metadate CD and Ritalin LA may be preferable for those ADHD patients who lose their appetite for dinner or have insomnia with Concerta

- Some patients may benefit from an occasional addition of 5–10 mg of immediate-release methylphenidate to their daily base of sustained-release methylphenidate
- Transdermal formulation may confer lower abuse potential than oral formulations
- Transdermal formulation may enhance adherence to treatment compared to some oral formulations because it allows once daily application with all day efficacy, has a smoother absorption curve,

and allows for daily customization of treatment (i.e., it can be removed early if desired)

- On the other hand, transdermal formulation has slower onset than oral formulations, requires a specific removal time, can cause skin sensitization, can be large depending on dose, and may lead to reduced efficacy if removed prematurely

Suggested Reading

Challman TD , Lipsky JJ . Methylphenidate: its pharmacology and uses .
Mayo Clin Proc 2000 ;**75** :711–21 .

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Kimko HC , Cross JT , Abemethy DR . Pharmacokinetics and clinical effectiveness of methylphenidate . *Clin Pharmacokinet* 1999 ;**37** :457–70 .

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Wolraich ML , Greenhill LL , Pelham W , et al. Randomized, controlled trial of oros methylphenidate once a day in children with attention-deficit/hyperactivity disorder . *Pediatrics* 2001 ;**108** :883–92 .

Mianserin

Therapeutics

Brands

- Lerivon
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: norepinephrine multimodal (N-MM)
- Tetracyclic antidepressant
- Noradrenergic agent

Commonly Prescribed for

(bold for FDA approved)

- Depression
- Anxiety
- Insomnia
- Treatment-resistant depression

How the Drug Works

- Blocks alpha 2 adrenergic presynaptic receptor, thereby increasing norepinephrine neurotransmission
- This is a novel mechanism independent of norepinephrine reuptake blockade
- Blocks alpha 2 adrenergic presynaptic receptors but also alpha 1 adrenergic receptors on serotonin neurons, thereby causing little increase in serotonin neurotransmission
- Blocks 5HT2A, 5HT2C, and 5HT3 serotonin receptors
- Blocks H1 histamine receptors

How Long Until It Works

- * Actions on insomnia and anxiety can start shortly after initiation of dosing
- Onset of therapeutic actions in depression, however, is usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6–8 weeks for depression, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine is stopped
- Continue treatment until all symptoms are gone (remission)
- Once symptoms gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders may also need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)

- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer

Best Augmenting Combos for Partial Response or Treatment Resistance

- SSRIs, SNRIs, bupropion, reboxetine, atomoxetine (use combinations of antidepressant with caution as this may activate bipolar disorder and suicidal ideation)
- Modafinil, especially for fatigue, sleepiness, and lack of concentration
- Mood stabilizers or atypical antipsychotics for bipolar depression, psychotic depression, or treatment-resistant depression
- Benzodiazepines

Tests

- Baseline ECG is recommended for patients over age 50
- May need to monitor blood count during treatment for those with blood dyscrasias, leukopenia, or granulocytopenia
- Since some antidepressants such as mianserin can be associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)

- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management
- * Monitor weight and BMI during treatment
- * While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antidepressant

Side Effects

How Drug Causes Side Effects

- Most side effects are immediate but often go away with time
- * Histamine 1 receptor antagonism may explain sedative effects
- * Histamine 1 receptor antagonism plus 5HT_{2C} antagonism may explain some aspects of weight gain

Notable Side Effects

- Sedation
- Increased appetite, weight gain



Life-Threatening or Dangerous Side Effects

- Rare seizures
- Rare blood dyscrasias
- Rare induction of mania
- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)

Weight Gain



- Many experience and/or can be significant in amount

Sedation



- Many experience and/or can be significant in amount
- Generally transient

What to Do About Side Effects

- Wait
- Wait
- Wait
- Switch to another drug

Best Augmenting Agents for Side Effects

- Often best to try another antidepressant monotherapy prior to resorting to augmentation strategies to treat side effects
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)
- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Many side effects cannot be improved with an augmenting agent
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of mianserin

Dosing and Use

Usual Dosage Range

- 30–60 mg/day

Dosage Forms

- Tablet 10 mg, 30 mg, 60 mg

How to Dose

- Initial 30 mg/day; maximum usually 90 mg/day



Dosing Tips

- Can be dosed once or twice daily
- If intolerable anxiety, insomnia, agitation, akathisia, or activation occur upon either dosing initiation or discontinuation, consider the possibility of activated bipolar disorder and switch to a mood stabilizer or an atypical antipsychotic

Overdose

- Relatively safe in overdose; sedation, hypertension, or hypotension

Long-Term Use

- Safe

Habit Forming

- Not expected

How to Stop

- Taper is prudent to avoid withdrawal effects, but tolerance, dependence, and withdrawal effects not reliably reported

Pharmacokinetics

- Half-life 12–29 hours



Drug Interactions

- Tramadol increases the risk of seizures in patients taking an antidepressant
- Carbamazepine and phenytoin may reduce mianserin levels
- Theoretically could cause a fatal “serotonin syndrome” when combined with MAOIs, so do not use with MAOIs or for at least 14 days after MAOIs are stopped unless you are an expert and only for treatment-resistant cases that may justify the risk
- Do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing mianserin

Other Warnings/Precautions

- Drug may lower white blood cell count (rare; may not be increased compared to other antidepressants but controlled studies lacking)
- Avoid alcohol, which may increase sedation and cognitive and motor effects
- Use with caution in patients with history of seizures
- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent
- When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient’s chart
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such

symptoms immediately

- Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient is taking an MAOI
- If there is a proven allergy to mianserin

Special Populations

Renal Impairment

- Dose adjustment not necessary

Hepatic Impairment

- Dose adjustment not necessary

Cardiac Impairment

- Baseline ECG is recommended
- Drug should be used with caution

Elderly

- Baseline ECG is recommended for patients over age 50
- Some patients may tolerate lower doses better